How to use these pages. They replace Section 1.5 (pages 7–9), the steering and BMS rows in Section 2 and add a short protection note. Edit the numbers back into your own CAD report; each block says which original text it replaces. Items marked **[verify]** must be checked against your exact part datasheets before submission.

1.5 Ramp Climbing Capability (replaces pages 7–9)

The rulebook gives the ramp heights (0.2, 0.4, 0.8 ft) but not their slope, so a single assumed angle is not enough. We therefore check **every slope from 5° to 30°** for the hardest ramp, h = 0.8 ft = 0.244 m, and state exactly which climbing method works where.

A. Torque required at the shell

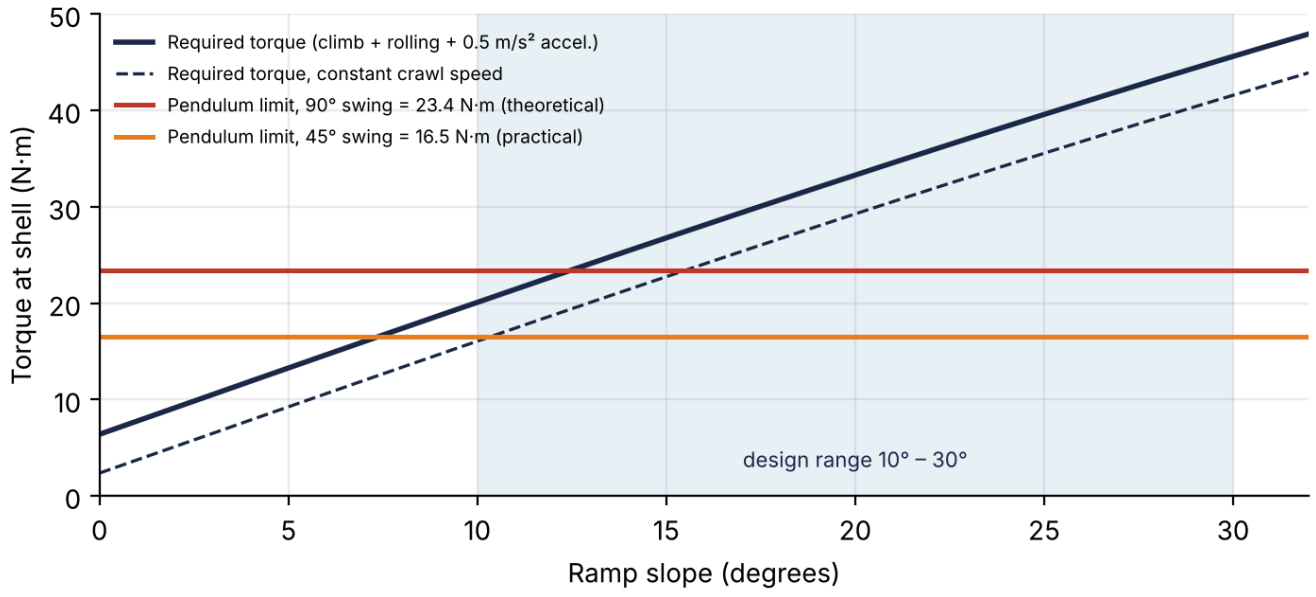
```
T_s   = m·g·R·sinθ                (hold the robot on the slope)
T_rr  = C_rr·m·g·R·cosθ           (rolling resistance, C_rr = 0.03)
T_a   = m·a·R                     (a = 0.5 m/s²)
T_req = T_s + T_rr + T_a

m = 24.8 kg, R = 0.325 m (over tread), g = 9.81 m/s²
```

B. Torque the pendulum body can react

The drive torque is reacted by the swung inner body, so it is limited by the body swing angle φ:

```
T_p(φ) = m_b·g·e·sinφ      m_b = 15.68 kg, e = 0.152 m
T_p(90°) = 23.4 N·m   (theoretical maximum)
T_p(45°) = 16.5 N·m   (practical: body swung 45° and still controllable)
```



Slope	T_s	T_rr	T_a	T_req	Crawl climb (no momentum) T_s+T_rr vs T_p
10°	13.7	2.3	4.0	20.1	**Yes** at 90° (16.0 ≤ 23.4); **Yes** at 45°, marginal (16.0 ≤ 16.5)

15°	20.5	2.3	4.0	26.8	**Yes** at 90° (22.8 ≤ 23.4); **No** at 45° (22.8 > 16.5)
20°	27.0	2.2	4.0	33.3	**No** – momentum needed
25°	33.4	2.1	4.0	39.6	**No** – momentum needed
30°	39.5	2.1	4.0	45.6	**No** – momentum needed

Result. Pure crawling works up to about **15.4°** with the full 90° swing (this is the earlier 15.5° figure) and about **10.3°** with the practical 45° swing. Above that the robot must carry speed into the ramp. **We therefore do not claim that the 0.2 ft and 0.4 ft ramps can be crawled**: that is true only if their slope is below these limits. All three ramps are taken with the same speed-assisted method below, which is valid for any slope up to 30°.

C. Speed-assisted climb (energy balance on the ramp)

The robot enters the ramp at speed v and keeps the drive pushing with the practical pendulum torque T_p(45°) all the way up. The ramp length is L = h / sinθ. The kinetic energy of the rolling shell is included through the inertia factor k, computed from the CAD shell mass instead of assumed:

```
k = (½·I_shell·ω²)/(½·m·v²) = (⅔·m_shell·R²/R²)/m = (⅔ × 9.12)/24.8 = 0.245
         (hollow shell of 9.12 kg; the inner body does not spin)

Work by drive on the ramp  W = ((T_p − T_rr) / R) · L
Energy needed             ΔE = m·g·h − W
Entry speed (20 % losses)  v = √( 2·ΔE / (m·(1+k)·0.8) )
```

Earlier we used k = 0.5. A larger k lowers the needed speed, so 0.5 was optimistic; the value from the real mass split is 0.245 and is used here.

Slope	Ramp length L (m)	Work by drive W (J)	m·g·h (J)	Entry speed needed v (m/s)	Margin vs 2.3 m/s
15°	0.94	41.3	59.4	1.21	1.9×
20°	0.71	31.5	59.4	1.50	1.5×
25°	0.58	25.6	59.4	1.65	1.4×
30°	0.49	21.7	59.4	1.75	1.3×

Smaller ramps need less: for h = 0.4 ft (0.122 m) and 0.2 ft (0.061 m) at 20° the entry speeds are 1.06 m/s and 0.75 m/s. The robot's dash speed under load is about 2.3 m/s (3.1 m/s no-load), so **all three ramps are covered at 30° with at least 1.3× speed margin**. Between the ramps the robot re-accelerates; the 20 m sector is long enough to regain 2.3 m/s from standstill at a = 0.5 m/s² ($v^2 = 2 \cdot a \cdot s = 20$ m²/s² over 20 m).

D. Motor and traction check

```
At the pendulum limit, shell torque 23.4 N·m is shared by 2 motors:
T_motor = 23.4 / (2 × 19 × 4 × 0.75) = 0.21 N·m   (≈ 21 % of the 1.0 N·m stall torque)

Traction limit: μ = 0.6  →  θ_traction = atan(0.6) = 31°
```

The motors are therefore never the limit; the pendulum is. Motor current at this torque is about a fifth of stall current, so it stays inside the MDD10A 10 A continuous rating **[verify with the 775 motor stall current]**. The earlier 114 N·m “stall torque” remains in the motor table only as an upper bound; **it is not used for the climb**.

At 30° the tread friction (0.6) is only just enough, so we keep the design range at 30° and flag traction as the next limit.

2.2 / 2.3 Steering motor supply (fixes the 12 V motor on a 22.2 V bus)

Problem

The JGY-370 steering gearmotor is a 12 V motor but the MD10C driver was fed from the 22.2 V battery rail, and the only rails on the MCB were 5.1 V, 5.0 V and 6.0 V.

Fix

Add a dedicated **12 V steering rail**: one more buck converter (fourth) taking the battery rail through the same 5 A fused branch, rated at least 5 A output and at least 40 V input **[verify]**, e.g. a Pololu D36V50F12-class module. The MD10C and the JGY-370 then run from 12 V and never see more than their rating, even if the firmware has a fault. As a second layer, the STM32 also limits steering PWM duty to **12 V / V_batt = 48 %** at a full 25.2 V pack.

Rail	Source	Feeds
22.2 V motor branch (22.2–25.2 V)	Battery → XT90-S key → 60 A fuse → contactor	MDD10A (drive motors), contactor coil (24 V)
12 V steering rail (new)	Buck 22.2 → 12 V, ≥ 5 A	MD10C and JGY-370
5.1 V	Buck	Raspberry Pi 5 + AI HAT+
5.0 V	Buck	Logic / MCB
6.0 V	Buck	Camera gimbal servo

Update the Section 2.2 “Logic branch” row to read: “5 A blade fuse, TVS, INA226 shunt, then four buck converters giving 5.1 V, 5.0 V, 6.0 V and 12 V”. Update Fig. 8 by adding one buck box between the INA226 and MD10C labelled “12 V”. In the motor table keep the JGY-370 row and add “Supply: 12 V rail”. The steering torque check (3.9 N·m needed, 4.9 N·m available) is unchanged.

2.2 Over-voltage protection (fixes the TVS and buck-converter gap)

Problem

Part	Rating (typical datasheet values) [verify]
6S LiPo full charge	25.2 V
SMBJ28A TVS	Stand-off 28 V, breakdown 31.1–34.4 V, clamps at ≈ 45 V
MP1584EN buck	Input 4.5–28 V
MDD10A and MD10C drivers	Input 5–30 V

A 28 V TVS does not start to conduct until above 31 V and lets a spike up to about 45 V reach the buck converters. Spikes are likely when the contactor opens under motor current (kill switch, over-current trip) because of wiring inductance. The TVS therefore does not protect the converters.

Fix

1. **Choose converters that tolerate the TVS clamp level.** Replace the MP1584EN modules with buck converters rated for at least 40 V input (50 V class preferred) **[verify]**. This is the key change; the TVS can only be trusted to limit spikes to about 45 V.
2. **Keep a TVS in the logic branch** (SMBJ28A is acceptable) to bound the energy of long spikes, and add a TVS of at least 600 W at the MCB input.
3. **Protect the drivers separately, since they are only rated 30 V.** Place at least 2 × 1000 µF, 35 V low-ESR capacitors directly at the MDD10A and MD10C supply terminals and keep the battery-to-driver leads short and twisted. The drive-motor back-EMF at full speed is about 24 V, so the bus stays below 30 V after the contactor opens.
4. **Open the contactor with the motors already commanded to zero** for every software-triggered stop (low cell, heartbeat loss, over-current). Only the hardware kill switch opens it under load.
5. **Charger:** set the balance charger to a hard 4.20 V per cell (25.2 V pack) limit, never above.

> The capacitor values and the spike level should be checked with a quick LTspice run of the contactor-opening event and written in the Round 2 simulation report; the numbers above are design values, not measured results.